

Karan Gupta

Full Stack Developer | Software Engineer

+91 9417618987 | guptakaran0720@gmail.com

Portfolio | GitHub | LinkedIn

Projects

EventFlow – Distributed Workflow Orchestration Engine

Tech: FastAPI, Python, PostgreSQL, Redis Streams, gRPC, Docker, WebSockets, SQLAlchemy, Next.js

GitHub: github.com/guptakaran20/EventFlow | Live: goeventflow.vercel.app

- Built a distributed workflow orchestration engine with custom workers, retries, fault recovery, and Dead Letter Queue (DLQ) support.
- Designed an event-driven architecture using Redis Streams, worker heartbeats, stale job recovery, and gRPC-based internal worker communication.
- Developed REST APIs, real-time WebSocket updates, Prometheus metrics, OpenTelemetry instrumentation, and PostgreSQL persistence with Docker deployment.

CodeArena – Real-Time Coding Battle Platform

Tech: Next.js, TypeScript, Node.js, Express.js, MongoDB, Redis, Socket.IO, Docker, AWS EC2, Nginx

GitHub: github.com/guptakaran20/CodeBattle | Live: codearenabattle.vercel.app

- Built a real-time competitive coding platform with battle rooms, live leaderboards, and synchronized multiplayer contests.
- Developed secure REST APIs using JWT authentication and Google OAuth with Redis and Socket.IO for matchmaking, timers, and live submissions.
- Integrated the Piston API for multi-language code execution and deployed on AWS EC2 using Docker Compose and Nginx.

ImportlyAI – AI-Powered CSV Import Platform

Tech: Next.js, TypeScript, Node.js, Express.js, Gemini API, PapaParse

GitHub: github.com/guptakaran20/ImportlyAI | Live: importlyai.vercel.app

- Built an AI-powered CSV importer that maps arbitrary CSV files into a standardized CRM schema using Gemini.
- Engineered a multi-stage extraction pipeline with schema inference, validation, normalization, and intelligent field mapping.
- Developed an automated benchmark suite with 180+ test cases and optimized batched inference for large, inconsistent datasets.

SponsorGrid – Sponsorship Management Platform

Tech: Next.js, TypeScript, Express.js, PostgreSQL, Prisma ORM, Cloudinary

GitHub: github.com/guptakaran20/Sponsorship | Live: trysponsorgrid.vercel.app

- Developed a full-stack SaaS platform for managing sponsors, partnerships, leads, and funding workflows.
- Designed PostgreSQL schemas with Prisma ORM and implemented Google OAuth, Cloudinary integration, and responsive dashboards using Next.js.

Technical Skills

Languages: C++, TypeScript, JavaScript, Python, SQL

Frontend: React.js, Next.js, Tailwind CSS, HTML5, CSS3

Backend: Node.js, Express.js, REST APIs, gRPC, FastApi, Socket.IO, WebSockets, JWT Authentication, Google OAuth

Databases: PostgreSQL, MongoDB, MySQL, Redis, Prisma ORM

DevOps & Tools: Docker, Docker Compose, AWS EC2, Nginx, Git, GitHub, Vercel, CI/CD

Experience

Backend Developer Intern

Jun 2026 – Present

XCEED NIT Jalandhar

- Contributing to the backend development of an Intelligent Attendance Management System (iAMS).

Education

Dr. B. R. Ambedkar National Institute of Technology, Jalandhar

B.Tech – Instrumentation and Control Engineering

2025 – 2029

CGPA: 8.49

Achievements

- Solved 500+ Data Structures and Algorithms problems on LeetCode Achieved a 1700 rating and a (2★) on CodeChef.
- Built and deployed multiple production-ready full-stack and distributed backend systems using FastAPI, Docker, Redis Streams, PostgreSQL, AWS EC2, and Nginx.
- Strong understanding of distributed systems, workflow orchestration, REST API design, Redis Streams, fault tolerance, and real-time communication.